

MATH3881/5881: Statistical Machine Learning Theory

Week 0: Basics of Probability Theory

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Key topics

- Random Variables
- Expectation, Variance, and Standard Deviation
- Confidence Intervals
- Divergences and Entropies
- Computations for Multivariate Normal Distributions

Reference:

- *Introduction to Probability* by Joseph K. Blitzstein & Jessica Hwang [click here for a pdf copy]
- Appendix B of *Mathematical Engineering of Deep Learning* by Lique, Moka, and Nazarathy: Freely available at <https://deeplearningmath.org/>

0.3.1 Random Variables

Set Theory Basics

- **Sample Space (Ω):** The set of all possible outcomes of an experiment.
- **Event:** A subset of Ω (e.g., $A \subseteq \Omega$).
- **σ -Algebra ($\mathcal{F}$):** A collection of events containing Ω and closed under complements, countable unions, and intersections.
- **Random Variable:** A function $X : \Omega \rightarrow \mathbb{R}$ is a random variable if $\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$ for all $x \in \mathbb{R}$.

Discrete Random Variables

- **Definition:** X takes countable values (e.g., integers), denote them by $\mathcal{X} \subset \mathbb{R}$.
- **Probability Mass Function (PMF):** $p_X(x) = P(X = x)$ for $x \in \mathcal{X}$.
- **Examples:**
 - Bernoulli: $p_X(1) = p, p_X(0) = 1 - p$.
 - Binomial: $p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}$.

Continuous Random Variables

- **Definition:** X takes uncountably infinite values (e.g., real numbers).
- **Probability Density Function (PDF):** $f_X(x)$ satisfies

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$

- **Examples:**
 - Uniform: $f_X(x) = \frac{1}{b-a}$ for $x \in [a, b]$.
 - Normal: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in \mathbb{R}$.

Cumulative Distribution Function (CDF)

- **Definition:** $F_X(x) = P(X \leq x)$, for any random variable X (discrete or continuous).
- **Properties:**
 - Non-decreasing: $F_X(x) \leq F_X(x')$ for all $x \leq x'$.

- Right-continuous: $\lim_{y \downarrow x} F_X(y) = F_X(x)$, $y \downarrow x$ denotes that y approaches x from the right (i.e., $y \rightarrow x^+$).
- $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$.
- For discrete X : $F_X(x) = \sum_{k \leq x} p_X(k)$.
- For continuous X : $F_X(x) = \int_{-\infty}^x f_X(t) dt$.

Expectation

- **Definition:**

- Discrete: $\mathbb{E}[X] = \sum_x x \cdot p_X(x)$.
- Continuous: $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$.

- **Linearity:** $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$.

- **Law of the Unconscious Statistician (LOTUS):** For any function $g: \mathbb{R} \rightarrow \mathbb{R}$,

- Discrete: $\mathbb{E}[g(X)] = \sum_x g(x) p_X(x)$.
- Continuous: $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$.

Variance and Standard Deviation

- **Variance:** $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

- **Standard Deviation:** $\sigma_X = \sqrt{\text{Var}(X)}$.

- **Properties:**

- $\text{Var}(aX + b) = a^2 \text{Var}(X)$.
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$.

Sample Mean and Sample Variance Estimators

- **Sample Mean:** $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.

- Unbiased: $\mathbb{E}[\bar{X}_n] = \mu$.

- **Sample Variance:** $S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.

- Unbiased: $\mathbb{E}[S_n^2] = \sigma^2$.

Confidence Interval (CI)

- **Definition:** An interval estimate for a parameter (e.g., μ) with a confidence level $(1 - \alpha)$.
- **For μ (Known σ):** CI is given by

$$\bar{X}_n \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} = \left(\bar{X}_n - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right),$$

where $z_{\alpha/2}$ is the $(1 - \alpha/2)$ quantile of $\mathcal{N}(0, 1)$.

- **For μ (Unknown σ):** CI is given by

$$\bar{X}_n \pm t_{\alpha/2, n-1} \frac{S_n}{\sqrt{n}} := \left(\bar{X}_n - t_{\alpha/2, n-1} \frac{S_n}{\sqrt{n}}, \bar{X}_n + t_{\alpha/2, n-1} \frac{S_n}{\sqrt{n}} \right),$$

where $t_{\alpha/2, n-1}$ is the quantile of the t -distribution with $n - 1$ degrees of freedom.

0.3.2 Divergences and Entropies

KL-Divergence for Discrete Distributions

- **Definition:** For discrete distributions $p(x)$ and $q(x)$ with supports $\mathcal{X}_p = \{x : p(x) > 0\}$ and $\mathcal{X}_q = \{x : q(x) > 0\}$:

$$D_{\text{KL}}(p \parallel q) = \sum_{x \in \mathcal{X}_p} p(x) \log \frac{p(x)}{q(x)}. \quad (0.1)$$

– If $\mathcal{X}_p \not\subseteq \mathcal{X}_q$, $D_{\text{KL}}(p \parallel q) = +\infty$.

- **Decomposition:**

$$D_{\text{KL}}(p \parallel q) = H(p, q) - H(p),$$

where:

- **Cross Entropy:**

$$H(p, q) = - \sum_{x \in \mathcal{X}_p} p(x) \log q(x). \quad (0.2)$$

- **Entropy:**

$$H(p) = - \sum_{x \in \mathcal{X}_p} p(x) \log p(x). \quad (0.3)$$

- **Binary Case:**

- Entropy: $H(p) = -(p_1 \log p_1 + (1 - p_1) \log(1 - p_1))$.
- Cross Entropy: $H(p, q) = -(p_1 \log q_1 + (1 - p_1) \log(1 - q_1))$.

- **Properties:**

- $D_{\text{KL}}(p \parallel q) \geq 0$ with equality iff $p = q$.
- Asymmetric: In general, $D_{\text{KL}}(p \parallel q) \neq D_{\text{KL}}(q \parallel p)$.

KL-Divergence for Continuous Distributions

- **Definition:** For continuous densities $p(x)$ and $q(x)$:

$$D_{\text{KL}}(p \parallel q) = \int_{\mathcal{X}_p} p(x) \log \frac{p(x)}{q(x)} dx. \quad (0.4)$$

Jensen-Shannon Divergence

- **Definition:** Symmetric divergence for $p(x)$ and $q(x)$ with supports $\mathcal{X}_p$ and $\mathcal{X}_q$:

$$\text{JSD}(p \parallel q) = \frac{1}{2} (D_{\text{KL}}(p \parallel m) + D_{\text{KL}}(q \parallel m)), \quad (0.5)$$

where $m(x) = \frac{1}{2}(p(x) + q(x))$.

– $\sqrt{\text{JSD}(p \parallel q)}$ is a valid metric.

0.3.3 Computations for Multivariate Normal Distributions

Multivariate Normal Density

- **PDF:** For $x \in \mathbb{R}^m$ with mean μ and covariance Σ :

$$\mathcal{N}(x; \mu, \Sigma) = \frac{1}{(\det \Sigma)^{1/2} (2\pi)^{m/2}} e^{-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)}.$$

- **Log-Density:**

$$\log \mathcal{N}(x; \mu, \Sigma) = -\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu) - \frac{m}{2} \log(2\pi) - \frac{1}{2} \log(\det \Sigma). \quad (0.6)$$

KL-Divergence for Multivariate Normals

- **General Case:** For $\mathcal{N}_{\mu_1, \Sigma_1}$ and $\mathcal{N}_{\mu_2, \Sigma_2}$:

$$D_{\text{KL}}(\mathcal{N}_{\mu_1, \Sigma_1} \parallel \mathcal{N}_{\mu_2, \Sigma_2}) = \frac{1}{2} \left((\mu_1 - \mu_2)^\top \Sigma_2^{-1} (\mu_1 - \mu_2) - m + \text{tr}(\Sigma_2^{-1} \Sigma_1) + \log \frac{\det \Sigma_2}{\det \Sigma_1} \right). \quad (0.7)$$

- **Special Cases:**

– For $\Sigma_2 = \sigma_2^2 I$:

$$D_{\text{KL}}(\mathcal{N}_{\mu_1, \Sigma_1} \parallel \mathcal{N}_{\mu_2, \sigma_2^2 I}) = \frac{1}{2\sigma_2^2} \|\mu_1 - \mu_2\|^2 + \frac{\text{tr}(\Sigma_1)}{2\sigma_2^2} - \frac{m}{2} + \frac{m \log \sigma_2^2}{2} - \frac{\log \det \Sigma_1}{2}. \quad (0.8)$$

– For standard normal ($\mu_2 = 0$, $\Sigma_2 = I$):

$$D_{\text{KL}}(\mathcal{N}_{\mu_1, \Sigma_1} \parallel \mathcal{N}_{0, I}) = \frac{1}{2} \|\mu_1\|^2 + \frac{\text{tr}(\Sigma_1)}{2} - \frac{m}{2} - \frac{\log \det \Sigma_1}{2}. \quad (0.9)$$